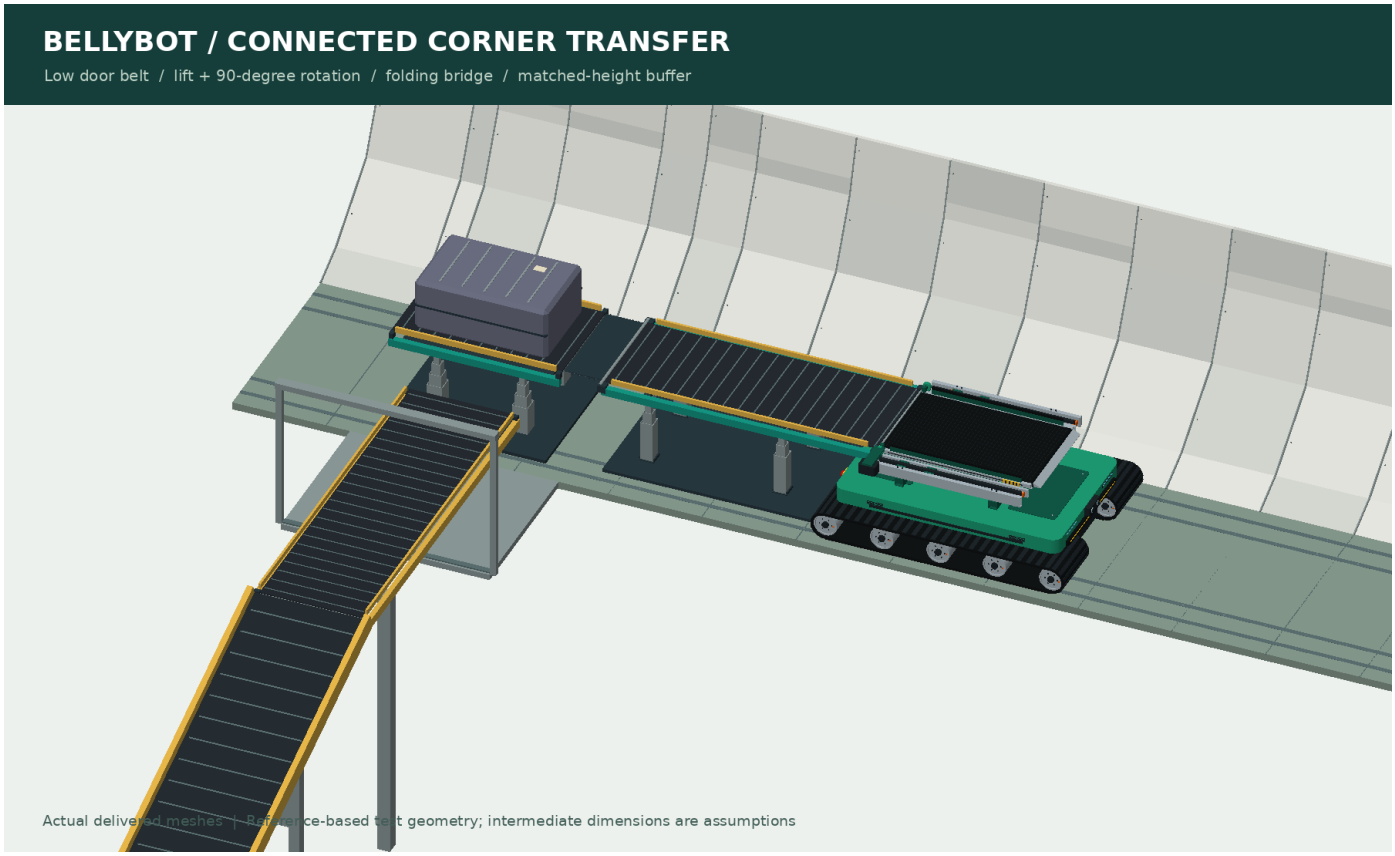
BELLYBOT
Door entry & corner transfer

REVISION 2 / 10 OCTOBER 2026 / CAD AND INTEGRATION REVIEW

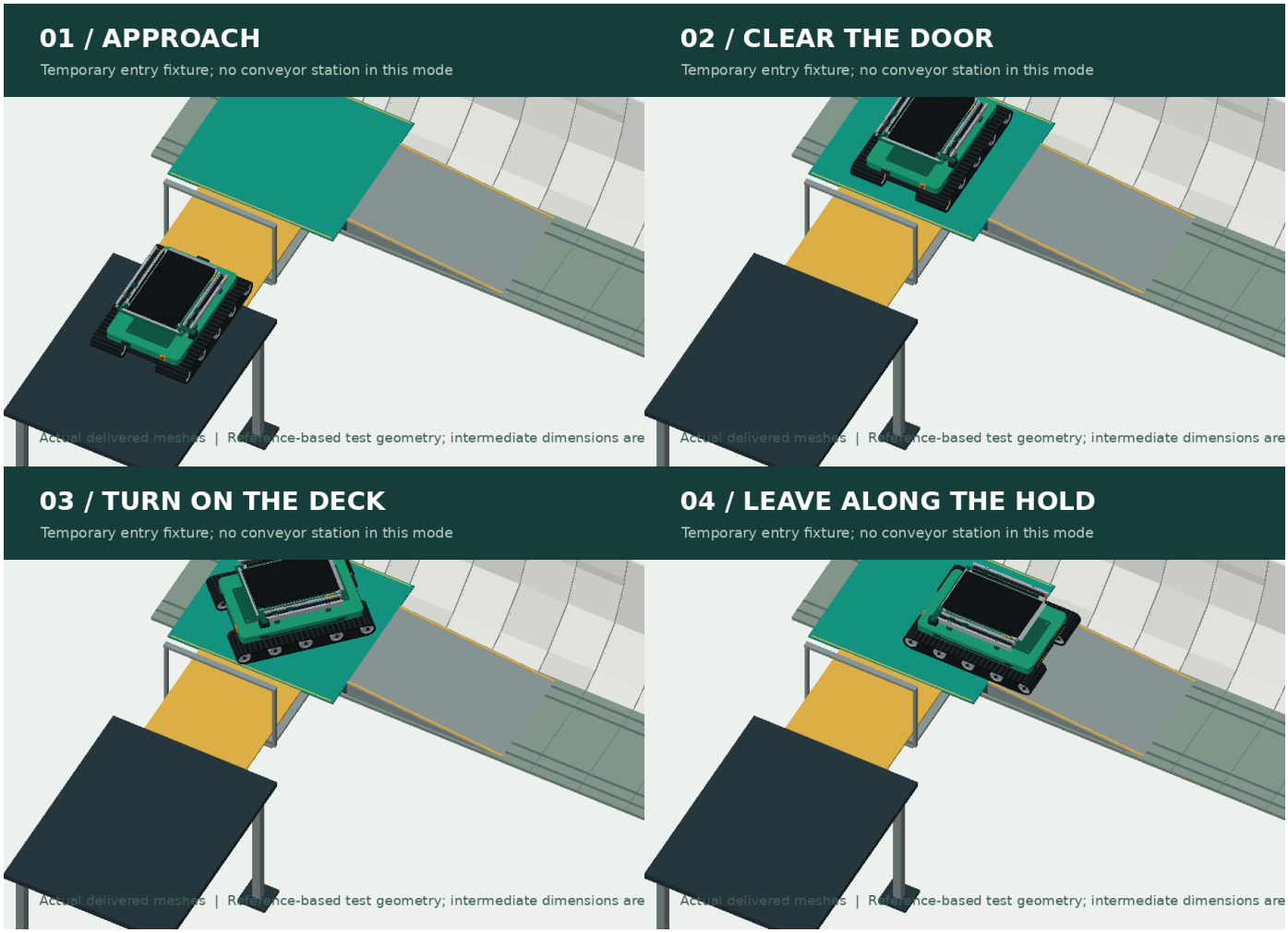


The robot fits in its stowed configuration. The original 1.10 m-wide crawler and telescopic tray are preserved. A temporary raised deck resolves the modeled door turn, and a lift-and-rotate table connects the side belt to the longitudinal receiving tray. The aircraft enclosure has not been enlarged.

Change	Delivered result
Door entry	Docked platform, 180 mm turn deck, exit ramp and feedback-based drive sequence.
Corner transfer	Low inlet belt, powered rotary table, folding bridge, supported buffer and joint feedback interlocks.
Operating limits	Controller-enforced height/pose checks and a chassis lock in work mode.
Team information	Team / lead / members / institute: complete before submission.

Verification boundary: CAD and offline geometry/logic checks passed. Gazebo/RViz, load/contact dynamics, 12 bags/min and aircraft-floor compliance remain unverified.

01 / A supported door-entry route



Entry property	Measured / specified value
Door opening	1.22 m wide x 0.89 m high; published narrow-body baseline [1].
Stowed robot on turn deck	About 0.718 m high above cargo floor; 172 mm nominal header clearance.
Jamb clearance	60.0 mm per side on the aligned approach.
Turning clearance	32.77 mm minimum in the modeled 90-degree pivot; >30.7 mm nominal continuous-path bound.
Exit ramp	1.80 m long, 1.18 m wide, 180 mm fall: 5.71 degrees.

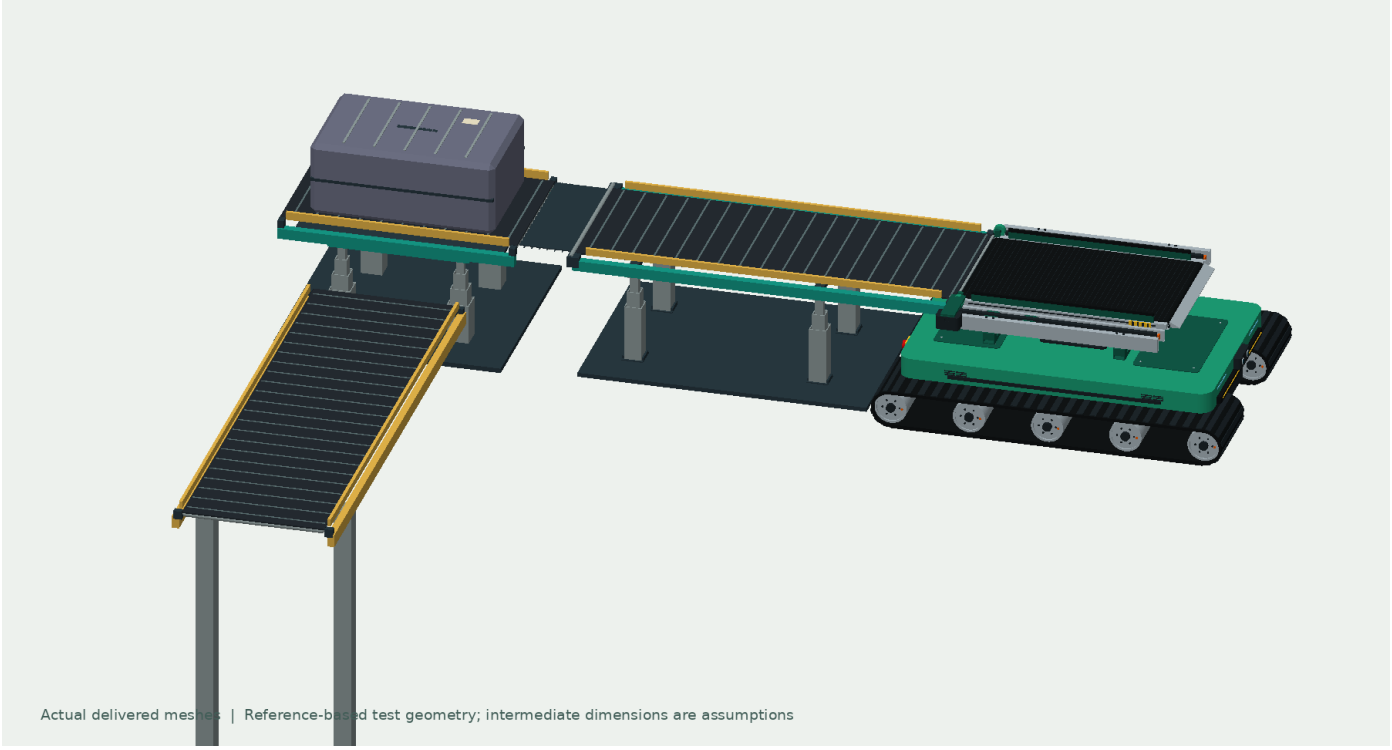
Enter empty with all robot joints stowed. Drive to the deck centre, turn the tracked base, then drive along the ramp to X=4.90 m. The external platform is assumed already docked and elevated. Floor contact and pitching are left to Gazebo physics; the driver does not teleport the base.

Entry and work use different fixtures. The temporary deck/ramp must be removed before the corner-transfer station is installed. Fixture installation/changeover and ramp stability are not automated or demonstrated in this delivery.

02 / The corner is an articulated mechanism

TRANSFER / MECHANICAL LAYOUT

0.300 m door belt -> 0.498 m receiving tray / rigid bodies remain physically separate



Actual delivered meshes | Reference-based test geometry; intermediate dimensions are assumptions

Component	Design and sequence
Level doorway belt	Surface Z=0.300 m; 0.60 m wide. The inclined loader ends 1.06 m outside the doorway so the bag levels before passing the header.
Lift-and-rotate table	1.00 x 0.60 m deck; three 113.33 mm lift stages and a bounded 90-degree joint. Stop and centre the bag, lift clear of the inlet, then rotate.
Folding bridge	228 mm powered nose folds downward before rotation; unfolds only after measured height and yaw alignment.
Longitudinal buffer	1.60 m belt. Raises to the robot receiving height and lowers to 0.370 m for upper-arm clearance. The original v3 infeed is replaced in these worlds.
Transfer gaps	Nominal 12 mm inlet gap, 8 mm bridge gaps and 2 mm final plate gaps. Soft bags and straps require contact trials.

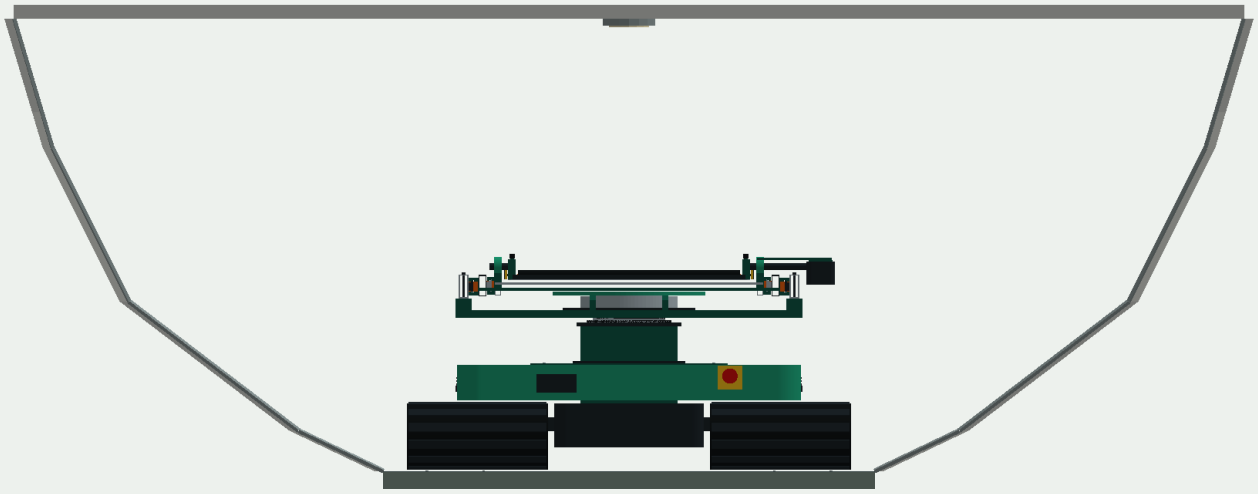
The rotor has a bearing, retained telescope guides and a supported load path. The URDF uses serial joints without a closed kinematic loop. Belts are contact-driven; their end rollers rotate about their shafts. Bags remain free rigid bodies. Industrial lift-and-rotate conveyors establish the mechanism class [2]; these custom fixtures are not an off-the-shelf rated product.

03 / What the operating limits mean

A joint can have a valid mechanical stroke that is too large for a particular aircraft. Here, the 600 mm mast stroke would put the unloaded robot about 18 mm through the modeled roof. The controller uses a shorter, bag-dependent operating stroke. That restriction preserves useful handling motion inside the hold.

CROSS SECTION / STOWED BELLYBOT

1.22 m flat floor / 1.10 m crawler width / no full chassis pivot assumed



Actual delivered meshes | Reference-based test geometry; intermediate dimensions are assumptions

Payload / motion	Enforced commissioning envelope
380 mm-high bag	130 mm total mast lift, leaving about 110 mm roof clearance.
320 mm-high bag	190 mm total mast lift, leaving about 110 mm roof clearance.
280 mm-high bag	230 mm total mast lift, leaving about 110 mm roof clearance.
Receiving from station	Maximum 130 mm mast lift for any bag; higher placement lifts follow receipt.
Upper yaw / reach	Yaw within ±65 degrees with lift >=80 mm and reach <=500 mm. Straight reach may reach 1 m if the whole-pose guard passes.
Chassis	Locked at X=4.10, Y=0 during work. Full chassis yaw is permitted only on the temporary entry deck.

The controller checks the original robot mesh hulls, the conservative bag envelope, 100 mm roof / 25 mm liner margins, and a four-patch assigned-mass pressure screen. Its allowed workspace does not include the separate aft-taper stress fixture or unmodeled piles.

04 / Verification and hackathon requirements

Evidence	Result and boundary
Door / turn	2,432 whole-mesh FCL samples; finite door jambs and liner panels are checked, including mesh edges.
Ramp	901 supported-height envelope poses; no tracked-vehicle stability or friction simulation.
Transfer mechanism	253 lift/rotation/bridge samples. Minimum deliberate bridge-interface separation about 7.97 mm.
Robot / parked buffer	477 sampled arm poses; conservative separation bound 46.0 mm.
Controller	Rejects excessive lift, low yaw, excessive yawed reach, lost bag pose, unfolded bridge, height mismatch and phase timeouts. Pure logic tests use synthetic feedback.
CAD / files	Entry and corner STEP assemblies reimport as valid solids; original robot STEP and SDF/URDF/meshes included. Local resources and positive inertia checks are recorded separately.

The hackathon performance requirements remain test gates. The 20–32 kg payloads are modeled, not demonstrated by selected hardware. The commissioning sequence is serial and slow; it does not achieve or establish 12 bags/min. A production cycle needs measured optimization, buffering/overlap and potentially a continuous corner conveyor.

The 600 kg/m² limit is not proven by whole-floor average or the assigned-mass controller calculation. Measure patch forces and actual contact areas. Anchored/static fixture models also bypass aircraft support reactions; their load-spreading structure and attachments need separate verification.

Submission deliverable [3]	Included / outstanding
STEP model	Original robot, aircraft fixture, entry dock and corner-transfer CAD.
MP4 <=2 minutes	Included: 30-second CAD motion preview. A final recorded Gazebo loading test is still needed.
5–8 page report	This six-page review; complete team details and measured results before submission.

Mesh clearance is evidence of geometric feasibility in the stated baseline, not certification of a specific aircraft. Intermediate wall slopes, the doorway pocket and fixture dimensions remain editable design assumptions.

05 / Start the integration from the revised launch

Build both packages in **ros2/**, then use **aircraft.launch.py**. It starts paused and wires the aircraft controller, real joint feedback and selected-bag pose feedback. Do not run the old v3 supervisor at the same time.

```
ros2 launch bellybot_aircraft_environment
aircraft.launch.py mode:=entry

ros2 topic pub --once /aircraft/start_entry
std_msgs/msg/Empty '{}'
```

For the separate work setup, use **mode:=transfer**. Unpause and spawn one free bag using **tools/spawn_transfer_bag.py --execute**, then publish **/aircraft/start_transfer**. The supplied adapter reads the simulator pose of **tracked_bag**; general baggage perception is not implemented.

Important file	Purpose
docs/AGENT_HANDOFF.md	Geometry, frame conventions, topic contracts, changeover scope and host commissioning sequence.
config/bellybot_operational_limits.yaml	Readable summary of the limits implemented in the controller.
scripts/aircraft_guard.py	Mesh-hull pose evaluation, height caps and mass/pressure rejection.
scripts/transfer_core.py	Measured-feedback transfer state machine and interlocks.
docs/resolution_checks.json	Numerical evidence and explicit runtime limitations.
tools/score_events.py	Score unique stable placements using simulation time; exclude drops and duplicates.

First host tests: joint signs and belt direction; slow empty entry and ramp settling; one bag of each geometry; pose-loss/jam stops; load-pressure measurements; then a timed repeated-loading run. RViz is a visualization of the articulated model, not a physics or safety test.

Sources

[1] Boeing, 737NG passenger overview, PDF p.17: boeing.com/content/dam/boeing/v2/company/startupboeing/historical/737NG_passenger.pdf.
[2] Dorner, Rotating Conveyors: dornerconveyors.com/solutions/rotating-conveyors.
[3] Supplied Hackathon Guidelines (Oct 2), included in reference/.
[4] Gazebo Sim 8 TrackController / PosePublisher and official ros_gz_bridge documentation. Full links: REFERENCES.md.
The user photographs inform the interior appearance. Video playback was unavailable; no observed video timing or mechanism detail is asserted.